A Appendix: Proofs of Results

A.1 Proof of Theorem 4.1

Proof. Let $\Gamma_n^\star$ and $\widehat{\Gamma}^\star$ denote the population and empirical Gram matrices after the normalization specified in (4.17). By the assumed local-support and regularity properties of the spline basis, together with the boundedness conditions on the density of X, the stipulated Gram-matrix concentration conditions yield

$$\left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| = o_{\mathbb{P}}(1)$$

whenever $K_n \log n/n \to 0$. In particular, under the corresponding concentration bound, one may write

$$\left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| = O_{\mathbb{P}}\left(\sqrt{\frac{K_n \log n}{n}}\right) = o_{\mathbb{P}}(1).$$

This establishes (4.21).

By (4.20), there exist constants $0 < c_0 < C_0 < \infty$ such that

$$c_0 \le \lambda_{\min}(\Gamma^\star) \le \lambda_{\max}(\Gamma^\star) \le C_0.$$

Weyl's eigenvalue inequality gives

$$\left|\lambda_{\min}(\widehat{\Gamma}^\star) - \lambda_{\min}(\Gamma^\star)\right| \le \left\|\widehat{\Gamma}^\star - \Gamma^\star\right\|$$

and

$$\left|\lambda_{\max}(\widehat{\Gamma}^\star) - \lambda_{\max}(\Gamma^\star)\right| \le \left\|\widehat{\Gamma}^\star - \Gamma^\star\right\|.$$

Consequently, with probability tending to one,

$$\lambda_{\min}(\widehat{\Gamma}^\star) \ge \frac{c_0}{2}, \qquad \lambda_{\max}(\widehat{\Gamma}^\star) \le C_0 + \frac{c_0}{2}.$$

Taking $c = c_0/2$ proves (4.22).

Finally, the normalization in (4.17) relates $\widehat{\Gamma}^\star$ to the original empirical Gram matrix $\widehat{\Gamma}$ through an invertible scaling transformation. Thus, nonsingularity of $\widehat{\Gamma}^\star$ implies nonsingularity of $\widehat{\Gamma}$. Hence,

$$\Pr\left\{\widehat{\Gamma} \text{ is nonsingular}\right\} \longrightarrow 1,$$

which proves (4.23). □

A.2 Proof of Theorem 5.1

Proof. Write

$$\widetilde{Z}_i^{\,0} = Z_i - \mu_Z(X_i), \qquad \Omega = E\left[\widetilde{Z}^{\,0}(\widetilde{Z}^{\,0})^\top\right].$$

By the assumed denominator condition (5.17), the normalized fitted-residual cross-product matrix satisfies

$$\frac{1}{n}\sum_{i=1}^{n} \widetilde{Z}_i \widetilde{Z}_i^\top \xrightarrow{\mathbb{P}} \Omega.$$

Since Ω is positive definite, the continuous mapping theorem implies

$$\left(\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i\widetilde{Z}_i^\top\right)^{-1} \xrightarrow{\mathbb{P}} \Omega^{-1}.$$

Using the model

$$Y_i = AZ_i + m(X_i) + \varepsilon_i$$

and the definition of the profile estimator, we obtain

$$\widehat{A} - A = \left\{\frac{1}{n}\sum_{i=1}^{n}\widetilde{Y}_i\widetilde{Z}_i^\top - A\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i\widetilde{Z}_i^\top\right\}\left(\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i\widetilde{Z}_i^\top\right)^{-1}.$$

The explicit same-sample remainder condition (5.23), together with the assumed denominator condition, therefore gives

$$\widehat{A} - A = \left\{\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(\widetilde{Z}_i^0)^\top\right\}\Omega^{-1} + o_{\mathbb{P}}(n^{-1/2}).$$

Here, the remainder condition is used directly to control the effect of estimating the conditional regression functions from the same observations; it is not inferred solely from convergence of the individual spline estimators.

Applying column-wise vectorization and the identity

$$\operatorname{vec}(BCD) = (D^\top \otimes B)\operatorname{vec}(C),$$

we obtain

$$\sqrt{n}\,\operatorname{vec}(\widehat{A} - A) = (\Omega^{-1} \otimes I_q)\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\operatorname{vec}\left[\varepsilon_i(\widetilde{Z}_i^0)^\top\right] + o_{\mathbb{P}}(1).$$

Define

$$U_i = \operatorname{vec}\left[\varepsilon_i(\widetilde{Z}_i^0)^\top\right] = \widetilde{Z}_i^0 \otimes \varepsilon_i.$$

The observations are independent and identically distributed, and $E(\varepsilon_i \mid X_i, Z_i) = 0$ implies $E(U_i) = 0$. Moreover, the assumed second-moment condition ensures that $E\|U_i\|^2 < \infty$. The multivariate central limit theorem therefore yields

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}U_i \xrightarrow{d} N(0, \Xi),$$

where

$$\Xi = \operatorname{Var}(U_i).$$

Under the stated conditional covariance assumption,

$$\Xi = E\left[\widetilde{Z}^0(\widetilde{Z}^0)^\top \otimes \Sigma(X)\right],$$

which agrees with the definition in (5.28).

It now follows from Slutsky's theorem that

$$\sqrt{n}\,\operatorname{vec}(\widehat{A} - A) \xrightarrow{d} N(0, \Sigma_A),$$

where

$$\Sigma_A = (\Omega^{-1} \otimes I_q)\Xi(\Omega^{-1} \otimes I_q)^\top.$$

Since Ω is symmetric, Ω^{-1} is symmetric as well, and the final factor can equivalently be written as $\Omega^{-1} \otimes I_q$. This proves (5.29)–(5.30). □

A.3 Proof of Theorem 6.1

Proof. Under Assumptions 6.1–6.6, the same-sample projection bounds (6.2)–(6.4) yield the asymptotic linear representation

$$\sqrt{n}\ \mathrm{vec}(\widehat{A} - A) = (\Omega^{-1} \otimes I_q)\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \mathrm{vec}\Big[\varepsilon_i(\widetilde{Z}_i^0)^\top\Big] + o_{\mathbb{P}}(1).$$

This is precisely (6.8).

Let

$$U_i = \mathrm{vec}\Big[\varepsilon_i(\widetilde{Z}_i^0)^\top\Big].$$

By the moment and conditional mean assumptions, $E(U_i) = 0$ and $E\|U_i\|^2 < \infty$. Furthermore,

$$\mathrm{Var}(U_i) = E\Big[\widetilde{Z}^0(\widetilde{Z}^0)^\top \otimes \Sigma(X)\Big] = \Xi.$$

The multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} U_i \xrightarrow{d} N(0, \Xi).$$

Applying Slutsky's theorem to the asymptotic linear representation establishes

$$\sqrt{n}\ \mathrm{vec}(\widehat{A} - A) \xrightarrow{d} N\big(0, (\Omega^{-1} \otimes I_q)\Xi(\Omega^{-1} \otimes I_q)^\top\big).$$

This proves (6.9) and (6.10), and completes the proof. □

A.4 Proof of Theorem 6.2

Proof. By the assumed functional weak convergence in (6.16),

$$G_n \Rightarrow G$$

in the function space specified in the theorem, where G is the mean-zero Gaussian process appearing in the limiting result. Define the functional

$$\Phi(f) = \int_{\mathcal{X}} \|f(x)\|^2 w(x)\, dx.$$

Under the stated conditions ensuring continuity of this functional on the relevant function space, the continuous mapping theorem gives

$$\Phi(G_n) \xrightarrow{d} \Phi(G).$$

Since T_n is the corresponding integrated squared-norm statistic, this yields

$$T_n \xrightarrow{d} \int_{\mathcal{X}} \|G(x)\|^2 w(x)\, dx.$$

It remains to express the limiting random variable in spectral form. Let $\mathcal{C}_w$ denote the covariance operator of G on the weighted function space associated with w. By the assumed trace-class condition, $\mathcal{C}_w$ is a positive, self-adjoint trace-class operator. Its

nonzero eigenvalues $\{\lambda_k\}_{k\geq 1}$ are nonnegative and summable, and the Karhunen–Loève representation gives

$$G = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, Z_k e_k,$$

in the corresponding mean-square sense, where $\{e_k\}_{k\geq 1}$ are orthonormal eigenfunctions and $Z_1, Z_2, \ldots$ are independent standard normal random variables. Consequently,

$$\int_{\mathcal{X}} \|G(x)\|^2 w(x)\, dx = \sum_{k=1}^{\infty} \lambda_k Z_k^2,$$

with convergence of the series almost surely and in mean. Therefore,

$$T_n \xrightarrow{d} \sum_{k=1}^{\infty} \lambda_k Z_k^2,$$

which proves (6.17). □